

Diabetes 130-US Hospitals – Data Cleaning Log

This document summarizes the data cleaning steps performed on the Diabetes 130-US Hospitals dataset.

Step	Column(s)	Cleaning Performed	Reason
1	encounter_id	Verified data type (int64), checked missing values and duplicate values.	Ensured the encounter identifier is complete and correctly stored.
2	patient_nbr	Verified data type (int64), checked missing values and duplicate values.	Confirmed patient identifier consistency.
3	race	Replaced '?' with NaN, filled missing values using the mode, and encoded categories into numerical values.	Standardized missing values and prepared the feature for analysis.
4	gender	Removed Unknown/Invalid records and encoded Female = 0, Male = 1.	Eliminated invalid categories and converted the column to numeric format.
5	age	Converted age ranges into midpoint numerical values (e.g., [30-40) → 35).	Made the age feature suitable for numerical analysis.
6	weight	Checked missing values and removed the column due to excessive missing data.	High proportion of missing values reduced its analytical usefulness.
7	admission_type_id	Verified data type, missing values, and unique values.	Confirmed the column was already stored correctly as numeric IDs.
8	discharge_disposition_id	Verified data type, missing values, and unique values.	Ensured data integrity before further analysis.
9	admission_source_id	Verified data type, missing values, and unique values.	Confirmed the column required no additional preprocessing.
10	diag_1, diag_2, diag_3	Replaced '?' with NaN and filled	Preserved records while standardizing

		missing values with 'Unknown'.	diagnosis information.
11	change	Encoded No = 0 and Ch = 1.	Converted the binary categorical feature into numerical format.
12	diabetesMed	Encoded No = 0 and Yes = 1.	Prepared the feature for analysis and modeling.
13	Medication Columns	Encoded medication status (No, Steady, Up, Down) into numerical values (0-3).	Converted categorical medication information into machine-readable format.
14	max_glu_serum	Encoded laboratory result categories into numerical values.	Preserved the order of glucose test results for analysis.
15	A1Cresult	Encoded laboratory result categories into numerical values.	Converted ordered categorical values into numeric levels.
16	readmitted	Encoded target labels (NO, >30, <30) as (0, 1, 2).	Prepared the target variable for predictive modeling.
17	Final Validation	Verified missing values, duplicate records, data types, dataset shape, and previewed the cleaned dataset.	Confirmed the dataset was clean and ready for exploratory analysis and machine learning.